



Grading differentiation in cutaneous squamous cell carcinoma: a review of the literature

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Abstract

Poor differentiation is strongly associated with poor outcomes in cutaneous squamous cell carcinoma (CSCC). In addition, the National Comprehensive Cancer Network (NCCN) guidelines designate poorly differentiated tumors as “very high risk”. Despite its clear prognostic implications, there is no standardized grading system for CSCC differentiation in common use today. CSCC differentiation is graded inconsistently by both dermatopathologists and Mohs surgeons, and reliability studies have demonstrated suboptimal inter- and intra-rater reliability in both of these groups. The absence of a standardized and reliable grading system has impeded the use of differentiation in CSCC staging, despite its apparent correlation with disease outcomes. We performed a comprehensive review of the literature summarizing historical CSCC differentiation grading systems, as well as grading systems in non-cutaneous head and neck SCC as a point of reference. Relevant articles were identified by searching Embase and PubMed, as well as by reviewing reference lists for additional articles and histology textbook excerpts. CSCC grading systems that were identified and summarized include the historical Broders system, the World Health Organization system, the College of American Pathologists’ system, and a system described by a 2023 Delphi consensus panel of dermatopathologists.

Keywords Cutaneous squamous cell carcinoma · Differentiation · Histologic grade · Staging · Outcomes

Introduction

Cutaneous squamous cell carcinoma (CSCC) is common, comprising up to 50% of skin cancer diagnoses in the United States [69]. Although most CSCCs are low-risk and curable with surgical intervention alone, a subset displays an aggressive phenotype and has a higher risk of poor outcomes including local recurrence, nodal metastasis, distant metastasis, and disease-specific death. High risk factors

that have been associated with poor outcomes include certain anatomic locations, immunosuppression, larger clinical diameter, greater depth of invasion, perineural invasion, and poor differentiation [13, 74].

In a 2016 meta-analysis of CSCC, poor differentiation was strongly associated with poor outcomes, with a 2.6-fold increase in local recurrence, a 5-fold increase in metastasis, and a 5.65-fold increase in disease specific death—the second strongest risk factor for this outcome [74]. In addition, the National Comprehensive Cancer Network (NCCN) guidelines designate poorly differentiated tumors as “very high risk” [71]. Despite its clear prognostic implications, there is no standardized grading system for CSCC differentiation in common use today. According to a survey of dermatopathologists, only 54% report the histologic grade of CSCCs on a consistent basis [43]. Similarly, a survey of Mohs micrographic surgeons demonstrated that only 70% consistently record CSCC differentiation, with reported barriers including lack of a standardized system, lack of reliability, and lack of training [59]. Furthermore, reliability studies have demonstrated suboptimal inter- and intra-rater

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reliability scores in grading CSCC differentiation in both Mohs surgeons and dermatopathologists [60, 66].

In 2018, the American Joint Committee on Cancer 8th edition (AJCC-8) removed histological differentiation as a CSCC staging criteria [3], although it remains a criterion in the Brigham and Women's Hospital staging system. Unfortunately, the absence of a standardized and reliable grading system has impeded the use of differentiation in CSCC staging, despite its apparent correlation with disease outcomes. Further research is needed to develop a standardized, reliable, user-friendly grading system for CSCC differentiation. To serve as a foundation for this future research, we performed a comprehensive review of the literature summarizing historical CSCC differentiation grading systems. We also performed a focused literature review of grading systems in non-cutaneous head and neck SCC (HNSCC) as a reference point, as differentiation grading has been fairly well studied in HNSCC.

Methods

An electronic literature search was conducted in Embase and PubMed using terms related to differentiation, histologic grade, and cutaneous squamous cell carcinoma, from database inception through July 10, 2023 (see Supplemental Material for the complete search strategies). Additional articles and relevant textbook chapters were identified by searching reference lists of included articles. Peer reviewed journal articles and textbook chapters that included information on grading histologic differentiation in cutaneous squamous cell carcinoma were reviewed and summarized. A second PubMed search was performed to identify articles on grading differentiation in non-cutaneous head and neck SCC; this included terms related to differentiation, histologic grade, and oropharyngeal SCC.

Literature review: differentiation grading systems in CSCC

Broders

The Broders system [16, 17] (Table 1) was used in the AJCC-7 staging manual [28], and was also presented in

many of the reviewed articles [6, 7, 14, 19, 20, 22–24, 31, 32, 35, 37, 38, 42, 43, 46, 47, 52, 55, 56, 61–63, 69, 76]. This system was first described in 1921 and grades CSCCs 1–4 based on the percentage of the tumor that is differentiated. Grade 1 tumors are 75% or more differentiated, Grade 2 tumors are 50–75% differentiated, Grade 3 tumors are 25–50% differentiated, and Grade 4 tumors are 0–25% differentiated. Broders did not specify a definition for “differentiated”, but noted that mitotic burden and the proportion of cells with large, dark, single nucleoli should be considered [16, 17]. Broders also stated that if there is difficulty choosing between 2 grades, to select the higher grade [17]. Broders' system has been criticized for its vagueness and lack of simplicity [23, 62].

Classically, Grade 1 tumors corresponded to well differentiated, Grade 2 to moderately differentiated, Grade 3 to poorly differentiated, and Grade 4 to undifferentiated [6, 42]. In 1990, Wick suggested in the textbook “Pathology of the Skin” that Broders Grade 1 be designated as well differentiated, Grades 2 and 3 (25–75% differentiated) be combined as moderately differentiated, and Grade 4 be considered poorly differentiated [77]; several articles utilized this modified grading [37, 46, 55, 62, 63]. Other articles combined Grades 1 and 2 (> 50% differentiation) as well differentiated [22, 56], and another grouped Grades 3 and 4 (< 50% differentiated) as poorly differentiated [35].

Bălăsoiu (2008) utilized the Broders system but defined differentiation as the presence of mature, keratinizing cells within tumor islands [7]. Kanoko (2000) utilized Broders' Grade 1–4 system, but offered specific definitions for each category; Grade 1 was defined as having prominent intercellular bridging and keratin pearl formation with mild anaplasia. Grade 2 was defined as having keratin pearl formation as well as single cell keratin production. Grade 3 was characterized by sparse single cell keratin production and significant anaplasia, and Grade 4 had significant anaplasia with no identifiable keratinization [42]. Weinberg (2007) suggested defining “differentiated” in the Broders system as areas with keratinization and horn pearls [76]. Ch'ng (2013) further expanded on the definition of “differentiated” in the Broders system as areas with prominent keratinization and horn pearls, a low level of pleomorphism, small nuclei with abundant cytoplasm, and strong cellular cohesion at the tumor borders [24]. McIlwee (2021) only considered keratinization in determining Broders grading [55].

World Health Organization (WHO)

Similar to the Wick modification of the Broders system, the WHO system for grading CSCC differentiation includes well differentiated (grade 1; <25% undifferentiated areas), moderately differentiated (grade 2; 25–75% undifferentiated

Table 1 Broders system for grading CSCC differentiation

Grade	Percent of differentiated cells in the tumor*
Grade 1	> 75%
Grade 2	50–75%
Grade 3	25–50%
Grade 4	< 25%

*Consider the mitotic burden and the proportion of cells with large, dark, single nucleoli

Table 2 World Health Organization (WHO) system for grading CSCC differentiation

Grade	Category	Percent of tumor with undifferentiated areas
Grade 1	Well differentiated	< 25% undifferentiated
Grade 2	Moderately differentiated	25–75% undifferentiated
Grade 3	Poorly differentiated	> 75% undifferentiated

Table 3 College of American Pathologists (CAP) system for grading CSCC differentiation

Category	Definition
Well differentiated	• Prominent keratinization • Prominent intercellular bridging • Low degree of pleomorphism • May have mitotic figures, primarily in the basal layer
Moderately differentiated	• Less prominent keratinization (typically restricted to individual keratin-filled cells and scattered keratin pearls and horn cysts) • Prominent pleomorphism • High mitotic burden
Poorly differentiated	• May have sparse keratinization or intercellular bridging • Difficult to identify as SCC
Undifferentiated	• Complete anaplasia

areas), and poorly differentiated (grade 3; >75% undifferentiated areas) [29]. See Table 2. Three of the reviewed articles utilized this system [27, 40, 57].

College of American Pathologists (CAP)

The CAP acknowledges that grading CSCC differentiation lacks standardization and reliability [72]. According to their published protocol, the Broders system can be used for grading [16]. Alternatively, CSCCs can be graded as well, moderately, poorly, or undifferentiated based on the degree of keratinization, cellular pleomorphism, intercellular bridges, and mitotic figures (Table 3) [21, 72]. In this system, well differentiated tumors have prominent keratinization and intercellular bridging, a low degree of pleomorphism, and may have mitotic figures but these should be primarily found in the basal layer. Moderately differentiated tumors have less prominent keratinization, typically restricted to individual cells and scattered pearls and cysts; in addition, there is a high mitotic burden and prominent pleomorphism. Poorly differentiated tumors may have sparse keratinization and/or intercellular bridging, but are overall difficult to identify as squamous, and undifferentiated tumors are completely anaplastic [21, 72]. Two other articles in this review utilized this grading system [11, 45].

Table 4 Delphi panel consensus guidelines for CSCC histology reporting

Category	Definition
Well differentiated	• High cytoplasm to nucleus ratio • Glassy atypia • Prominent keratinization
Moderately differentiated	• Some keratinization • No spindled or anaplastic cells
Poorly differentiated	• Sparse to absent keratinization • Spindled or anaplastic cells • Often require immunohistochemical stains to identify as SCC

CSCC histology consensus guidelines

In 2023, a group of dermatopathologists completed a Delphi process to determine consensus guidelines for reporting the histology of actinic keratoses and CSCCs (Table 4) [25]. A minor focus of this study was grading CSCC differentiation. The panel recommended a well, moderately, and poorly differentiated grading system; moving to 2 categories was considered, but consensus was not achieved. The panel characterized well differentiated tumors as having a high cytoplasm to nucleus ratio, “glassy atypia”, and keratinization. Moderately differentiated CSCCs were characterized as having some keratinization and no spindled or anaplastic cells, and poorly differentiated tumors as having spindled or anaplastic cells, sparse to absent keratinization, and often requiring immunohistochemical staining [25].

Histologic features considered in CSCC differentiation grading

When using a well, moderate, and poorly differentiated grading system, there is substantial variation in the literature in terms of which individual histologic features should be considered. The textbook “Lever’s Histopathology of the Skin” (10th edition) and other articles suggest grading CSCCs as well, moderately, or poorly differentiated based primarily on the degree of keratinization [49–51, 58, 69].

Other articles considered a wide variety of histologic features including keratinization [2, 4, 8, 10, 26, 30, 36, 41, 44, 48, 53, 64–68, 73, 75], cellular atypia [8, 30, 33, 53, 54, 64–68, 73, 75], pleomorphism [4, 8, 10, 41, 53, 64–66, 73], spindled cells [26, 66], mitotic Figs. [36, 44, 53, 64–67, 73], architecture [2, 8, 33, 48, 53, 64, 66], tumor edge growth pattern [2, 8, 48, 53, 64, 66, 67], intercellular bridging [26, 30, 53, 64, 67, 68], peritumoral stroma [4], and the need for IHC staining to identify the tumor [48]. Tumor staining pattern with H&E was also considered in 2 articles (noting that well differentiated tumors are eosinophilic and poorly differentiated tumors tend to be more basophilic) [36, 44].

Literature review: differentiation grading systems in non-cutaneous head and neck SCC

Articles on non-cutaneous head and neck SCC are also summarized here as a reference point. Histology grading systems for non-cutaneous head and neck SCC have been well studied, and a variety of systems have been described over the last century. As in cutaneous SCC, the first grading system was proposed by Broders in 1920 and was identical to the CSCC system described above [15]. In 1973, a grading system was proposed by Jakobsson in which the tumor shape (ranging from “papillary and solid” to “marked cellular dissociation”), degree of keratinization, nuclear pleomorphism, number of mitoses, tumor architecture (ranging from well circumscribed to a “diffuse growth” pattern), depth of invasion, vascular invasion, and lymphohistiocytic inflammatory response were each graded on a 1–4 scale. Total scores ranged from 8 to 32, with higher scores representing worse differentiation [39]. While this system was more comprehensive and correlated to poor outcomes [5], it was complex, definitions were vague, and it would have been fairly time consuming to use in clinical practice. This grading system was refined by Anneroth et al. in 1987 [5]. The new version kept the 1–4 scoring system, but only considered 6 parameters and clearly defined each; this included keratinization (1=over 50% keratinizing, 2=20–50%, 3=5–20%, 4=0–5%), pleomorphism as defined by the percentage of “mature” appearing cells (1=greater than 75%, 2=50–75%, 3=25–50%, 4=0–25%), number of mitotic figures per high powered field (1=0–1, 2=2–3, 3=4–5, 4=greater than 5), tumor architecture, depth of invasion, and inflammatory response [5]. While this system was slightly more simple and included more specific definitions for each parameter, it was still fairly complicated to use.

Later grading systems were designed to be simpler and more user friendly. The WHO grading system for head and neck SCC is based on the original Broders system and considers only keratinization, pleomorphism, and mitotic burden. Well differentiated tumors appear similar to normal squamous tissues, moderately differentiated tumors have less prominent keratinization, pleomorphic cells, and mitotic figures, and poorly differentiated tumors are mostly comprised of pleomorphic cells, with prominent mitotic figures and sparse keratinization [9]. This system has been criticized for being vague, subjective, and not accurately predicting poor disease outcomes [70]. A grading system proposed by Bryne evaluated 4 parameters (keratinization, pleomorphism, tumor growth pattern, and inflammatory response) at only the worst-appearing portion of the tumor [18]. Another system proposed in 2005 scored just 3 features (perineural involvement, inflammatory response at the tumor edges, and the worst-appearing invasion pattern) with

a 3-point scale [12]. Finally, a system called the “BD score” determined histologic grade on a 0–2 scale which included depth of invasion in millimeters and the degree of tumor budding. Tumor budding is defined as small clusters of 1–4 malignant cells at the deep aspect of the tumor. The overall scoring system was: 0=less than 4 mm depth and fewer than 5 tumor buds, 1=either greater than 4 mm depth or greater than 5 buds, and 2=both greater than 4 mm depth and greater than 5 buds [1]. In a study directly comparing these 4 grading systems, the BD score performed best at predicting poor disease outcomes [70].

Discussion

While poor differentiation in CSCC has been shown to be predictive of poor outcomes [74], there is no standardized, validated grading system in common use today. In addition, CSCC differentiation grading lacks reliability within and between raters [60, 66]. These issues have led to clinicians grading differentiation inconsistently [43, 59], as well as concerns with including differentiation in CSCC staging systems [3]. In the CSCC literature, there is a sparsity of articles describing differentiation grading; in addition, articles that do address this topic present a wide variety of grading methods, most of which are highly subjective. There is wide variability in which histologic features are considered, how they are defined and scored, and how they are incorporated into the overall grade. Even in non-cutaneous head and neck SCC, in which more work on this subject has been done, optimal differentiation grading remains controversial.

As poor differentiation is strongly correlated to poor CSCC outcomes, developing a reliable, valid, and user-friendly grading system is an important goal for improved CSCC prognostication. Historical grading systems have been criticized for either being overly complicated and not feasible to use in the clinical setting, or too simplistic, vague, and subjective. The ideal system would include histologic parameters which are predictive of poor CSCC outcomes, but are also quick and easy to grade.

One consideration for simplifying CSCC differentiation grading is a dichotomous system in which tumors are either high grade (poorly or undifferentiated) or low grade (moderately or highly differentiated). Historical CSCC grading systems – Broders numerical grading as well as the well/moderately/poorly/undifferentiated system – are all characterized by 3 or more categories. This likely negatively impacts precision and reliability, as it creates too much middle ground and allows graders to hedge. For CSCC prognostication and staging, the most important decision point is whether or not a tumor is poorly differentiated; thus, there may not be inherent value in having more than

2 categorizations. A dichotomous system would be easier to define, and may improve reliability. However, there are two major concerns with a dichotomous system. First, a dichotomous system has not yet been shown to correlate to poor CSCC outcomes. Second, moderate differentiation is an intermediate risk factor for poor outcomes, and a dichotomous grading system would fail to identify these tumors [34].

In conclusion, further research is needed to develop a standardized, reliable, user-friendly grading system for CSCC differentiation that is predictive of poor outcomes. This comprehensive review of the literature summarizing historical SCC differentiation grading systems will serve as a foundation for the development of new grading systems and future research on this topic.

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Declarations

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